

Solutions — IMC Training, May 13, 2026

Problem 1. (IMC 2022, Day 1, Problem 2) Real $n \times n$ matrices with real eigenvalues and $A + A^k = A^T$, $k \geq n$.

Solution. *Answer:* $A = 0$.

If λ is an eigenvalue of A , then $\lambda + \lambda^k$ is an eigenvalue of $A^T = A + A^k$, hence of A . Take λ with the largest absolute value: if k is odd, the map $\lambda \mapsto \lambda + \lambda^k$ strictly increases the absolute value of any nonzero λ , a contradiction unless every eigenvalue is 0. If k is even, the same conclusion follows by comparing traces of A^T and $A + A^k$.

Therefore A has only the eigenvalue 0 and is nilpotent; $A^n = 0$ and a fortiori $A^k = 0$. The equation then gives $A = A^T$, i.e. A is real symmetric with all eigenvalues zero, so $A = 0$. $\square$

Problem 2. (IMC 2021, Day 1, Problem 2) $\mathbb{P}(\min Y > \max X)$ is independent of a .

Solution. The pair (X, Y) is chosen out of $\binom{k+a}{k} \binom{n+k+a}{n}$ equally likely pairs.

When $\min Y > \max X$, the union $X \cup Y \subseteq \{1, \dots, k+n+a\}$ has size $n+k$ and the splitting between X and Y (smallest k elements vs. largest n) is determined uniquely. The condition forces every element of X to be $\leq k+a$, but among the $n+k$ chosen the smallest k are automatically $\leq n+k+a-n = k+a$. So the favorable count equals $\binom{n+k+a}{n+k}$.

Hence

$$\mathbb{P}(\min Y > \max X) = \frac{\binom{n+k+a}{n+k}}{\binom{k+a}{k} \binom{n+k+a}{n}} = \frac{1}{\binom{n+k}{k}},$$

where the last equality is a routine factorial cancellation. $\square$

Problem 3. (IMC 2020, Day 2, Problem 5) $f''f \geq 2(f')^2$, $f > 0$, $f \in C^2(\mathbb{R}) \Rightarrow f$ is constant.

Solution. Set $g = 1/f$. Then

$$g'' = \left(\frac{1}{f}\right)'' = \left(-\frac{f'}{f^2}\right)' = \frac{2(f')^2 - f''f}{f^3} \leq 0,$$

so g is a positive concave function on $\mathbb{R}$.

For any $a < b$, concavity gives, for all $u < a$ and $v > b$,

$$\frac{g(a) - g(u)}{a - u} \geq \frac{g(b) - g(a)}{b - a} \geq \frac{g(v) - g(b)}{v - b}.$$

Since $g(u), g(v) > 0$, the left side is $> \frac{-g(u)}{a-u} \cdot 0 = 0$ wait — more carefully: the left side equals $\frac{g(a)}{a-u} - \frac{g(u)}{a-u}$, and taking $u \rightarrow -\infty$ yields 0 (both terms $\rightarrow 0$). Similarly the right side tends to 0 as $v \rightarrow \infty$. The squeeze gives $g(a) = g(b)$, so g is constant and so is f .

Any positive constant function trivially satisfies the inequality. $\square$

Problem 4. (IMC 2022, Day 1, Problem 3) $f(p)$ = number of $(p-1)$ -step walks on $\mathbb{Z}$ with steps $\{-1, 0, 1\}$ returning to 0; find $f(p) \pmod{p}$.

Solution. *Answer:*

$$f(p) \equiv \begin{cases} 0 \pmod{3}, & p = 3, \\ 1 \pmod{p}, & p \equiv 1 \pmod{3}, \\ -1 \pmod{p}, & p \equiv 2 \pmod{3}. \end{cases}$$

$f(p)$ is the constant term of $(x+1+x^{-1})^{p-1}$, i.e. the coefficient $[x^{p-1}]$ of $(1+x+x^2)^{p-1}$. In $\mathbb{F}_p[[x]]$,

$$(1+x+x^2)^{p-1} = \frac{(1-x^3)^{p-1}}{(1-x)^{p-1}} = \frac{(1-x^3)^p(1-x)}{(1-x)^p(1-x^3)} \equiv \frac{(1-x^{3p})(1-x)}{(1-x^p)(1-x^3)} \pmod{p},$$

using $(1-y)^p \equiv 1-y^p \pmod{p}$. For $k = p-1 < p < 3p$, the factors $(1-x^{3p})$ and $(1-x^p)^{-1}$ do not affect the coefficient of x^{p-1} . So

$$f(p) \equiv [x^{p-1}] \frac{1-x}{1-x^3} = [x^{p-1}](1-x) \sum_{j \geq 0} x^{3j} \pmod{p}.$$

Write $p-1 = 3m+r$ with $r \in \{0, 1, 2\}$; the coefficient is 1 if $r = 0$, -1 if $r = 1$, 0 if $r = 2$. $\square$

Problem 5. (IMC 2020, Day 2, Problem 7) $[G : H_1] = [G : H_2] = n$, $[G : H_1 \cap H_2] = n(n-1) \Rightarrow H_1, H_2$ conjugate.

Solution. Set $K = H_1 \cap H_2$. From

$$n(n-1) = [G : K] = [G : H_1][H_1 : K] = n[H_1 : K]$$

we get $[H_1 : K] = n-1$. Write $H_1 = \bigsqcup_{i=1}^{n-1} h_i K$. Since $K \subseteq H_2$, each $h_i K \cdot H_2 = h_i H_2$, so

$$H_1 H_2 = \bigsqcup_{i=1}^{n-1} h_i H_2,$$

a disjoint union of $n-1$ left cosets of H_2 . (Disjointness: $h_i H_2 \cap h_j H_2 \neq \emptyset \Leftrightarrow h_i^{-1} h_j \in H_2$, but also in H_1 , hence in K , giving $h_i K = h_j K$.)

Therefore $L := G \setminus H_1 H_2$ is the remaining left coset of H_2 . By the symmetric argument L is also the remaining right coset of H_1 . So for any $g \in L$,

$$g H_2 = L = H_1 g, \quad \text{hence} \quad H_2 = g^{-1} H_1 g. \quad \square$$

Problem 6. (IMC 2021, Day 2, Problem 8) Maximum number of unit vectors in $\mathbb{R}^n$ with any 3 having 2 orthogonal.

Solution. *Answer:* $2n$, achieved by $\pm e_1, \dots, \pm e_n$.

For the upper bound, let $e_1, \dots, e_N$ be the unit vectors. The Gram matrix $A = (e_i \cdot e_j)$ is positive semidefinite of rank $\leq n$; equivalently $B := A - I$ has eigenvalue -1 of multiplicity at least $N - n$, and all other eigenvalues ≥ 0 .

The diagonal entries of B^3 are

$$(B^3)_{ii} = \sum_{j,k \neq i, j \neq k} a_{ij} a_{jk} a_{ki}.$$

The hypothesis means every summand has at least one zero factor (some pair among e_i, e_j, e_k is orthogonal), so $\text{trace}(B^3) = 0$.

Let P_i be the projection onto $\mathbb{R}e_i$. Then $B = \sum_i P_i - I$, so the condition $\text{trace}(B^3) = 0$ with $Q = \sum_i P_i$ and $\text{trace} Q = N$ gives, after expansion using $\text{trace}(P_i P_j P_k) = 0$ for distinct i, j, k and $\text{trace}(P_i^2 P_j) = \text{trace}(P_i P_j)$:

$$\sum t_i^3 = \text{trace}(Q^3) = 3 \sum t_i^2 - 2N,$$

where t_i are the eigenvalues of Q (there are at most n of them). Hence $\sum t_i^3 - 3t_i^2 = -2N$. But $(t-2)^2(t+1) = t^3 - 3t^2 + 4 \geq 0$ for $t \geq -1$ (which holds since $Q \succeq 0$), giving

$$-2N = \sum (t_i^3 - 3t_i^2) \geq -4n,$$

hence $N \leq 2n$. $\square$

Problem 7. (IMC 2022, Day 2, Problem 7) k idempotent matrices with $A_i A_j = -A_j A_i$ for $i \neq j$. Show some A_i has rank $\leq n/k$.

Solution. *Lemma.* For any idempotent B , $\text{rank}(B) = \text{trace}(B)$. (Proof: $B^2 = B$ means the minimal polynomial divides $\lambda(\lambda-1)$, so B is diagonalizable with eigenvalues 0, 1; the trace counts 1-eigenvalues, which equals the rank.)

Now compute $(\sum_i A_i)^2$:

$$\left(\sum_{i=1}^k A_i \right)^2 = \sum_i A_i^2 + \sum_{i \neq j} (A_i A_j + A_j A_i) = \sum_i A_i + 0 = \sum_i A_i,$$

so $\sum_i A_i$ is idempotent. Apply the lemma:

$$\sum_{i=1}^k \text{rank}(A_i) = \sum_{i=1}^k \text{trace}(A_i) = \text{trace}\left(\sum_i A_i\right) = \text{rank}\left(\sum_i A_i\right) \leq n.$$

So at least one $\text{rank}(A_i) \leq n/k$. $\square$

Problem 8. (IMC 2020, Day 2, Problem 8) $\lim_{n \rightarrow \infty} \frac{1}{\log \log n} \sum_{k=1}^n (-1)^k \binom{n}{k} \log k$.

Solution. *Answer:* 1.

Use the Frullani integral $\log k = \int_0^\infty \frac{e^{-x} - e^{-kx}}{x} dx$. Then

$$S_n := \sum_{k=1}^n (-1)^k \binom{n}{k} \log k = \int_0^\infty \frac{1}{x} \left(\sum_{k=1}^n (-1)^k \binom{n}{k} (e^{-x} - e^{-kx}) \right) dx.$$

The inner sum equals $-e^{-x} \cdot ((1-1)^n - 1) - ((1 - e^{-x})^n - 1) = -e^{-x} + 1 - (1 - e^{-x})^n$, so

$$S_n = \int_0^\infty \frac{1 - e^{-x} - (1 - e^{-x})^n}{x} dx.$$

Asymptotic estimate. Bernoulli gives $(1 - e^{-x})^n \geq 1 - ne^{-x}$, so the tail

$$\int_M^\infty \frac{1 - e^{-x} - (1 - e^{-x})^n}{x} dx \leq n \int_M^\infty \frac{e^{-x}}{x} dx \leq nM^{-1}e^{-M}.$$

Choose M with $Me^M = n$, so $M = \log n - \log \log n + o(1)$, and the tail is $O(1)$.

On $[0, M]$, uniformly $(1 - e^{-x})^{n-1} \leq e^{-(n-1)e^{-x}} \leq e^{-(n-1)M/n} \rightarrow 0$, so

$$\int_0^M \frac{1 - e^{-x} - (1 - e^{-x})^n}{x} dx = (1 + o(1)) \int_0^M \frac{1 - e^{-x}}{x} dx = (1 + o(1))(\log M + O(1)) = (1 + o(1)) \log \log n.$$

Therefore $S_n / \log \log n \rightarrow 1$. □

Problem 9. (IMC 2021, Day 2, Problem 7) $|f(0)| \leq \max_{|z|=1} |f(z)p(z)|$ for holomorphic f and monic polynomial p .

Solution. Let $p(z) = z^n + a_{n-1}z^{n-1} + \dots + a_0$ and define the *reciprocal polynomial*

$$q(z) = z^n \overline{p(1/\bar{z})} = 1 + \overline{a_{n-1}}z + \dots + \overline{a_0}z^n.$$

For $|z| = 1$, $1/\bar{z} = z$, so $|q(z)| = |\overline{p(z)}| = |p(z)|$. Also $q(0) = 1$ and q is holomorphic on $\mathbb{C}$, so $f \cdot q$ is holomorphic on a neighborhood of the closed unit disk. The maximum-modulus principle gives

$$|f(0)| = |f(0)q(0)| \leq \max_{|z|=1} |f(z)q(z)| = \max_{|z|=1} |f(z)p(z)|. \quad \square$$

Problem 10. (IMC 2020, Day 1, Problem 3) Centrally symmetric convex polytope approximation with $C(d)\varepsilon^{1-d}$ vertices.

Solution. Pick an inclusion-maximal set $\{p_1, \dots, p_m\} \subset \partial K$ such that the homothets $K_i := p_i + \frac{\varepsilon}{2}K$ have pairwise disjoint interiors. Set $L := \text{conv}\{p_1, \dots, p_m\}$.

Bound on m . Each $K_i \subseteq (1 + \frac{\varepsilon}{2})K$ by convexity of K . Also, if $a > 0$ and $ak \in K_i$ for some $k \in K$, then $p_i \in ak - \frac{\varepsilon}{2}K = ak + \frac{\varepsilon}{2}K$ (by central symmetry), so $p_i \in (a + \frac{\varepsilon}{2})K$; since $p_i \in \partial K$ we get $a \geq 1 - \frac{\varepsilon}{2}$. Hence all K_i lie in $(1 + \frac{\varepsilon}{2})K \setminus (1 - \frac{\varepsilon}{2})K$. Comparing volumes,

$$m \cdot (\varepsilon/2)^d \text{vol}(K) \leq ((1 + \frac{\varepsilon}{2})^d - (1 - \frac{\varepsilon}{2})^d) \text{vol}(K) \leq (3/2)^d \varepsilon,$$

so $m \leq 3^d \varepsilon^{1-d}$, and we may take $C(d) = 3^d$.

Inclusion $(1 - \varepsilon)K \subseteq L$. Suppose for contradiction that $p \in (1 - \varepsilon)K \setminus L$. Separate p from L by a linear functional ℓ : $\ell(p) > \max_i \ell(p_i)$. Let $x \in K$ attain $a := \max_{x \in K} \ell(x)$. By maximality of $\{p_i\}$, the homothet $x + \frac{\varepsilon}{2}K$ intersects some K_i : take $z \in (x + \frac{\varepsilon}{2}K) \cap (p_i + \frac{\varepsilon}{2}K)$. Then

$$\ell(p_i) + \frac{\varepsilon}{2}a \geq \ell(z) \geq \ell(x) - \frac{\varepsilon}{2}a = a - \frac{\varepsilon}{2}a,$$

so $\ell(p_i) \geq a(1 - \varepsilon)$. But $p \in (1 - \varepsilon)K$ gives $\ell(p) \leq a(1 - \varepsilon) \leq \ell(p_i)$, contradicting $\ell(p) > \ell(p_i)$. Hence $L \subseteq K$ (immediate) and $(1 - \varepsilon)K \subseteq L$. □